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AGRICULTURE
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TALSANDE

**D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY
TALSANDE**

MCA SANDWICH

PROJECT NAME

**VISUALIZING HOUSING MARKETS TRENDS: AN ANALYSIS OF SALE
PRICES AND FEATURES USING TABLEAU**

TEAM MEMBERS

- 1) SHREYA PAWAR**
- 2) SAKSHI PHAKADE**
- 3) SHIVANI BILE**
- 4) GAYATRI GAVADE**

1) Data Collection & Extraction from Database:

❖ Collect the dataset:

- 1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.

The screenshot shows the Skill Wallet dashboard. On the left is a blue sidebar with navigation links: Dashboard, Skill Bank, Skill Wallet (selected), Skill Card, and Skill Credits. The main content area has a breadcrumb trail: Dashboard > Skillwallet > Program > Data Analytics with Tableau ... > Group-projects > Visualizing Housing Market Tre... The title of the project is 'Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau'. Below the title, there's a section 'Collect the dataset' with a 'Story' button, a 'Duration: 2h 30m' badge, and a status 'Assigned to Shreya Pawar'. A 'Description' box contains the text: 'Acquire the finalized dataset required for the Tableau project, ensuring it is clean, relevant, and aligned with the defined problem. Validate data integrity and readiness for analysis and visualization tasks.' Under 'Resources', there is a 'Dataset Link' button. The 'Subtasks (1)' section shows a task 'Understand the data' with a 'Medium' difficulty level.

- 2) From the Kaggle Download the Dataset **Transformed_Housing_Data2.csv** by clicking on **Download** button,

The screenshot shows the Kaggle dataset page for 'Transformed Housing Data 2'. The page has a sidebar with navigation links: Create, Home, Competitions, Datasets (selected), Models, Benchmarks, Game Arena, Code, Discussions, Learn, More, Your Work, and VIEWED. The main content area shows the dataset title 'Transformed Housing Data 2' with a 'Download' button. Below the title, there's a 'Data Card' section with a 'Detail' tab selected. The 'About this file' section states: 'There are a total of 31 columns, out of which Sale Price can be supposedly taken as a dependent variable. The other variables are different features, locations and date, etc. regarding the houses.' Below this, there's a table with 6 columns: # Sale_Price, # No of Bedrooms, # No of Bathrooms, # Flat Area (in Sqft), # Lot Area (in Sqft), and # No of Houses. Each column has a brief description and a corresponding histogram. On the right, there's a 'Data Explorer' section showing 'Version 1 (2.41 MB)' and a 'Summary' section showing '1 file' and '31 columns'.

❖ Loading the Dataset:

Step 1: Click on “Microsoft Excel” or “Text File”

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)

Click the correct option.

Step 2: Select Your File

Browse your computer.

Select your dataset file.

Click Open.

Step 3: Data Source Page

Your sheet names will appear on the left side.

Drag the sheet to the centre area.

Tableau will show a preview of your data.

Tableau Desktop interface showing the Data Source page for 'Transformed_Housing_Data2'. The interface includes a top menu bar, a toolbar, and a sidebar with 'Connections' and 'Files' sections. The main area displays a preview of the CSV data with columns: Sale Price, No of Bedrooms, No of Bathrooms, Flat Area (in Sqft), and Lot Area (in Sqft). A 'Fields' pane on the left lists the available fields and their data types.

Name	Transformed_Housing_Data2.csv
Sale Price	221900.00
No of Bedrooms	3
No of Bathrooms	1.00000
Flat Area (in Sqft)	1180.00
Lot Area (in Sqft)	5.650.00
	538000.00
	3
	2.25000
	2.570.00
	7.242.00
	180000.00
	2
	1.00000
	770.00
	10.000.00
	604000.00
	4
	3.00000
	1.960.00
	5.000.00
	510000.00
	3
	2.00000
	1.680.00
	8.080.00
	1129575.00
	4
	4.50000
	5.420.00
	101.930.00
	257500.00
	3
	2.25000
	1.715.00
	6.819.00
	291850.00
	3
	1.50000
	1.060.00
	9.711.00
	229500.00
	3
	1.00000
	1.780.00
	7.470.00
	323000.00
	3
	2.50000
	1.890.00
	6.560.00

2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.

The screenshot displays the Tableau Public interface for a data source named "Transformed_Housing_Data2". The interface includes a sidebar with "Connections" and "Files" sections, a main view area displaying a table of house sale data, and a bottom navigation bar with tabs for "Data Source", "Sheet 1", "Sheet 2", "Sheet 3", "Sheet 4", "Sheet 5", "Dashboard 2", and "Story 1".

The main view area shows a table with the following columns:

Name	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv	Transformed_Housing_Data2.csv
Description	No description available.				
Fields					
Type	Field Name	Phys...	Rem...		
#	Sale Pric	Transf...	Sale...		
#	No of Be	Transf...	No of ...		
#	No of Ba	Transf...	No of ...		
#	Flat Area	Transf...	Flat Ar...		
#	Lot Area	Transf...	Lot Ar...		
#	No of Fl	Transf...	No of ...		

The table displays the following data rows:

Sale Price	No of Bedrooms	No of Bathrooms	Flat Area (in Sqft)	Lot Area (in Sqft)
221900.00	3	1.00000	1,180.00	5,650.00
538000.00	3	2.25000	2,570.00	7,242.00
180000.00	2	1.00000	770.00	10,000.00
604000.00	4	3.00000	1,960.00	5,000.00
510000.00	3	2.00000	1,680.00	8,080.00
1129575.00	4	4.50000	5,420.00	101,930.00
257500.00	3	2.25000	1,715.00	6,819.00
291850.00	3	1.50000	1,060.00	9,711.00
229500.00	3	1.00000	1,780.00	7,470.00
323000.00	3	2.50000	1,890.00	6,560.00

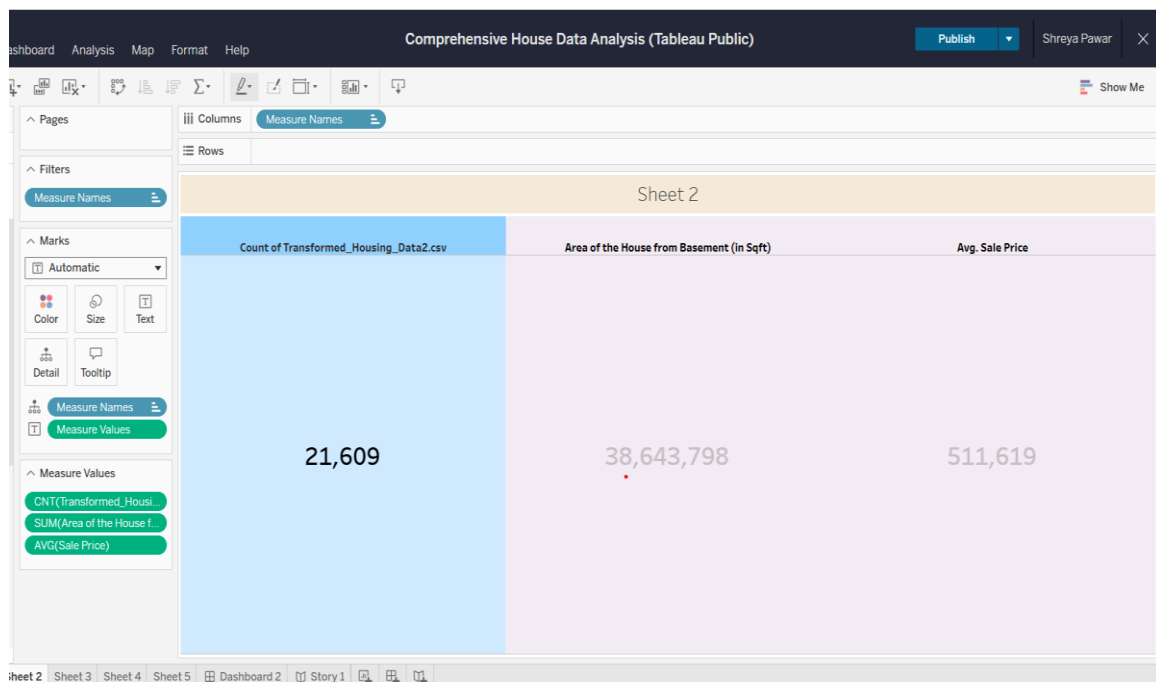
3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

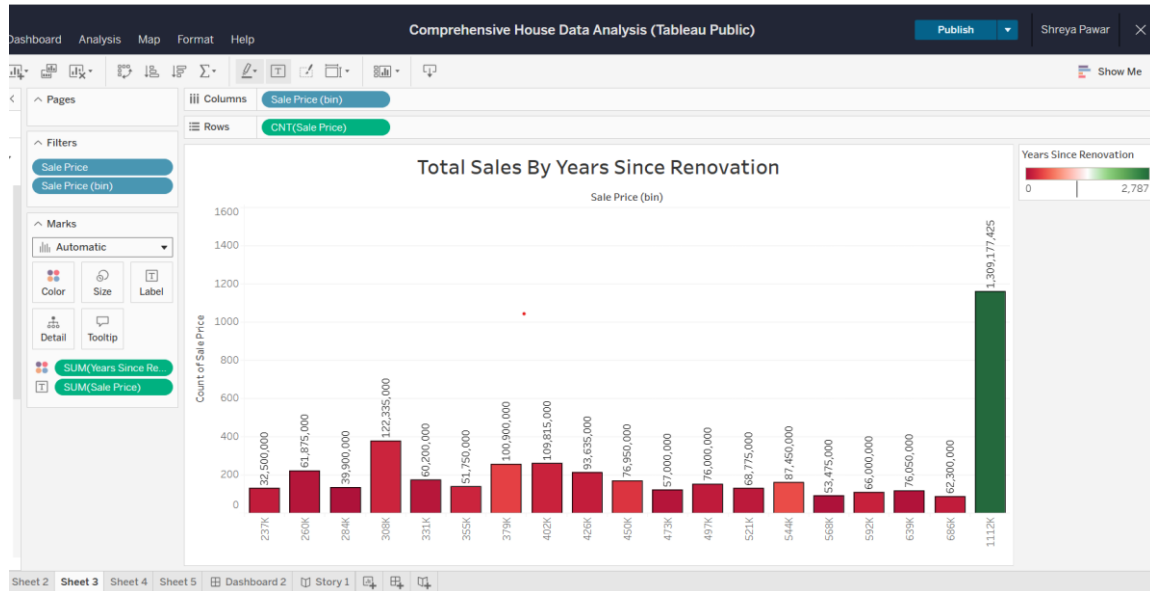
1) Count of Transformed_Housing_Data2.csv

- a) Drag and Drop the Measure Name and Measure Values.
- b) Add only-
 - Count of Transformed_Housing_Data2.csv
 - Area of the House from Basement (in Sqft)
 - Avg.SalePrice
- c) Drag and Drop Measure Name at Column section.



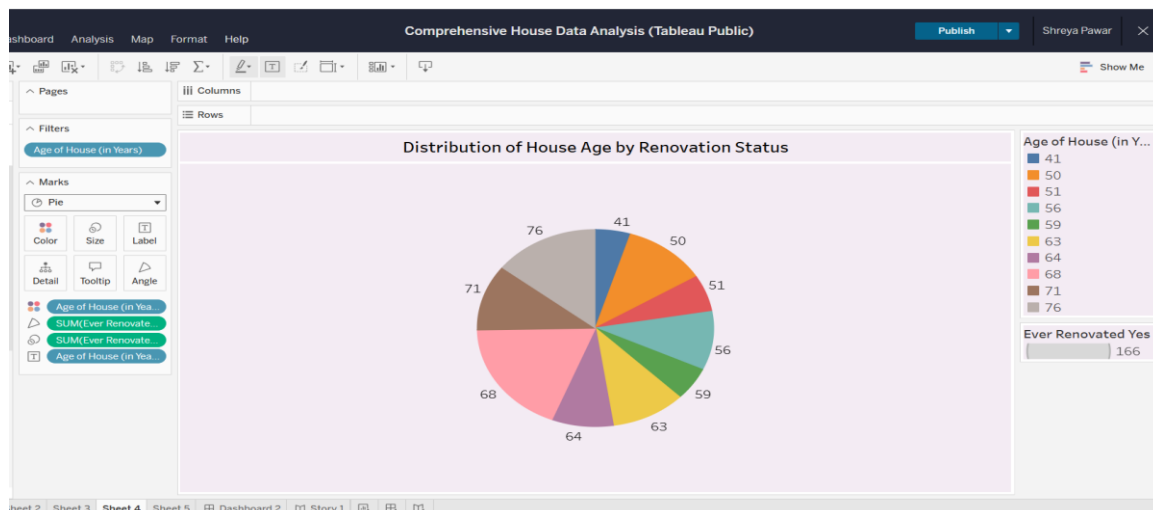
2) Total Sales by Years Since Renovation

- Convert the Sales Price into Dimension by right clicking.
- Add Filters into Sales Price and Sales Price(bin) to display Top 25 records.
- Drag and Drop Sales Price(bin) for grouping Sales Price into column section, and also CNT(Sale price) into rows.
- Drag and Drop SUM(Year Since Renovation) into color and SUM(Sale Price) into Label.



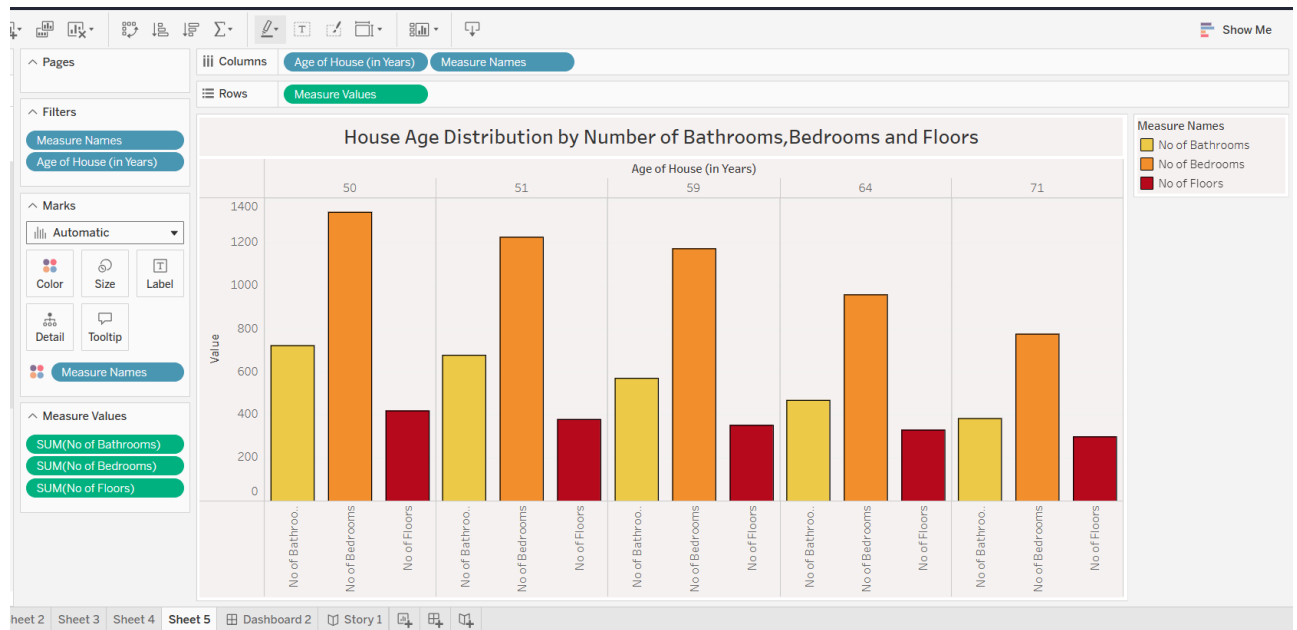
3) Distribution of House Age by Renovation Status

- Convert Age of House(in Years) to Dimension.
- Filter Age of House(in Years) into Top 10 records.
- Add Age of House(in Years) into color.
- Add SUM(Ever Renovated yes) into angle.
- Add Sum(Ever Renovated yes) into size.
- Add Age of House(in Years) in to label.



4) House Age Distribution by Number of Bathrooms, Bedrooms and Floors

- Drag and Drop Measure Name and Age of House(in Years).
- Filter Age of House(in Years) into Top 5 records.
- Drag and Drop Measure Values-
 - SUM(No of Bathroom)
 - SUM(No of Bedroom)
 - SUM(No of Floors)
- Drag and Drop Measure Names into Color.
- Add Age of House(in Year) and Measure Name into column section, and Measure Values into rows.



4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case.

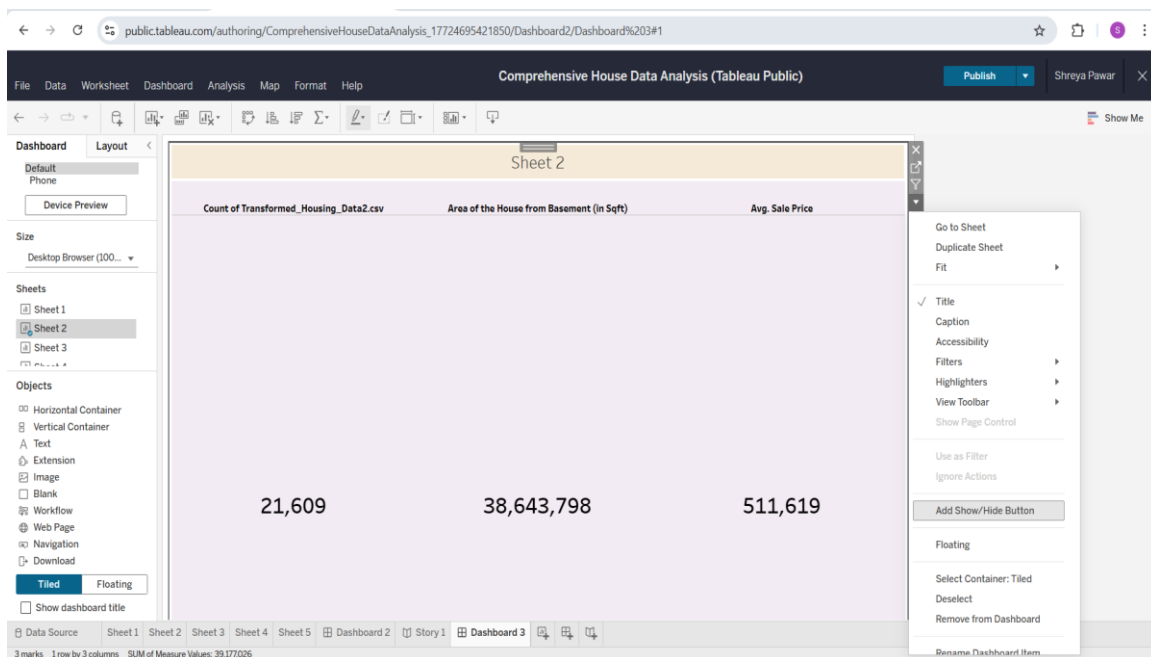
Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖ Responsive and Design of Dashboard

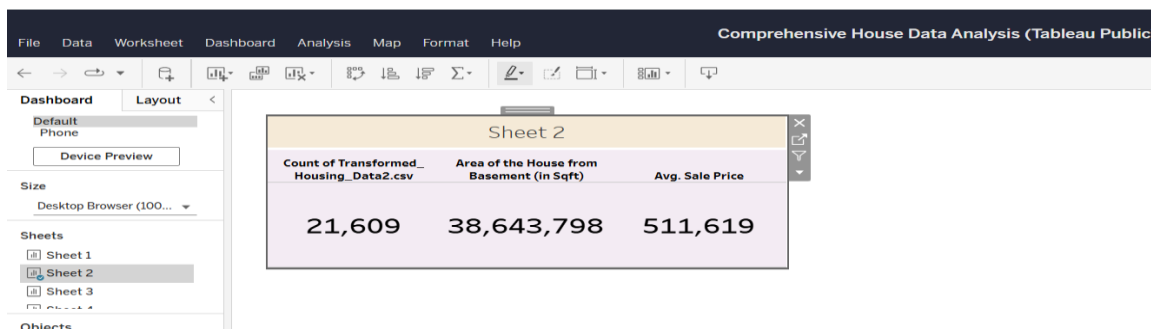
a) Click On **New Dashboard** option.



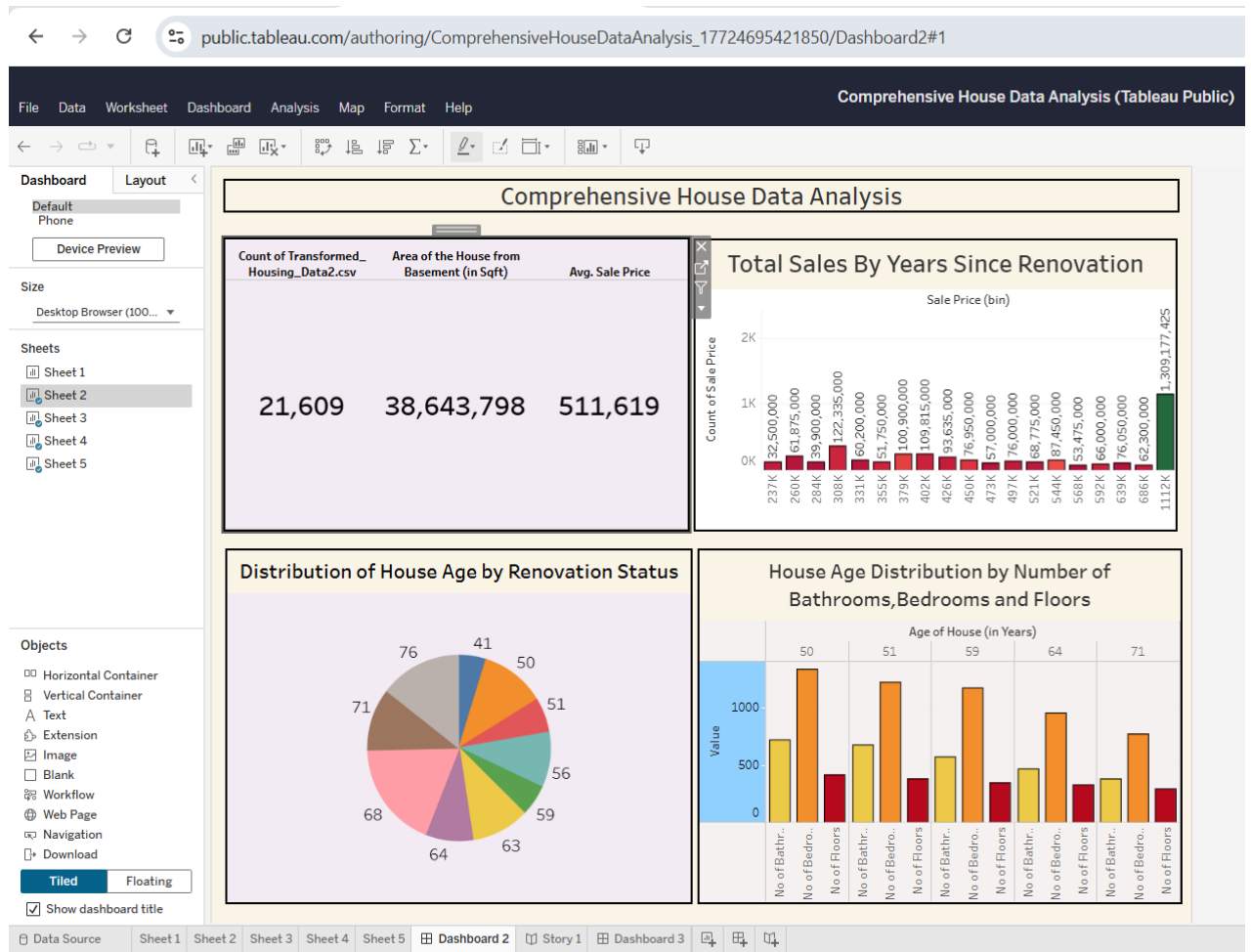
b) Drag and Drop Sheets from the data panel into centre white screen and in the right side of sheet there is Downward arrow so click on that and choose **Floating** option.



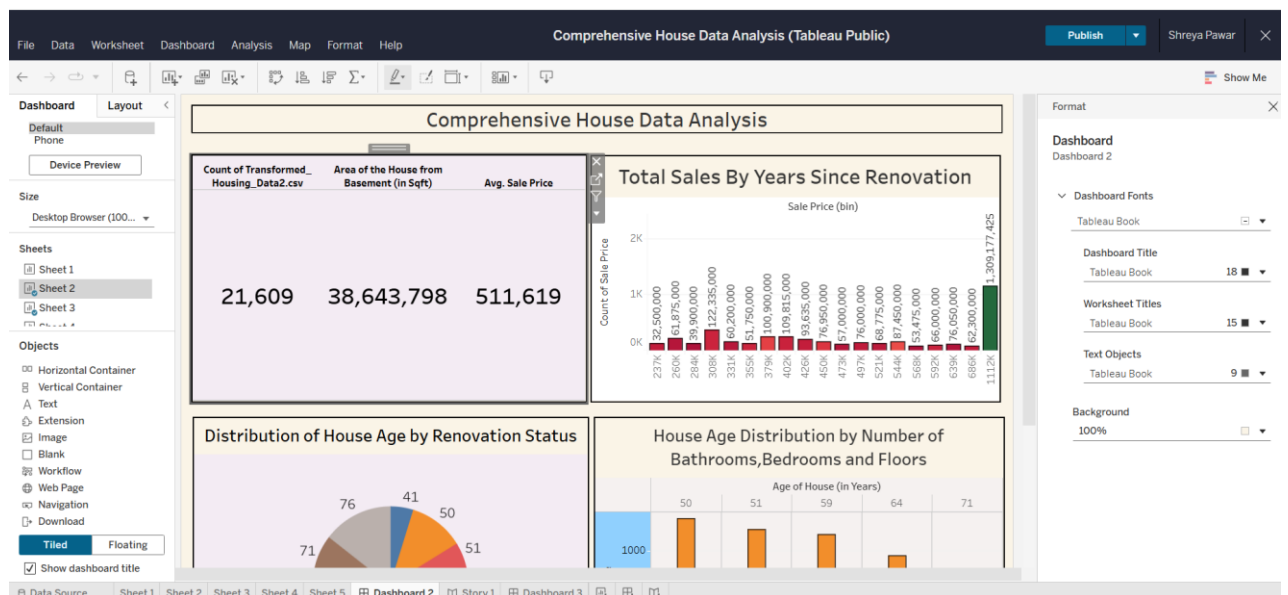
c) After selecting Floating option it's looks like.



d) Drag and Drop the Required Sheets for dashboard there I added 4 sheets.



e) Add a Format to an Dashboard by clicking at Format option at the Top and add background color to dashboard and boarder to the sheets.



5) Storyboard

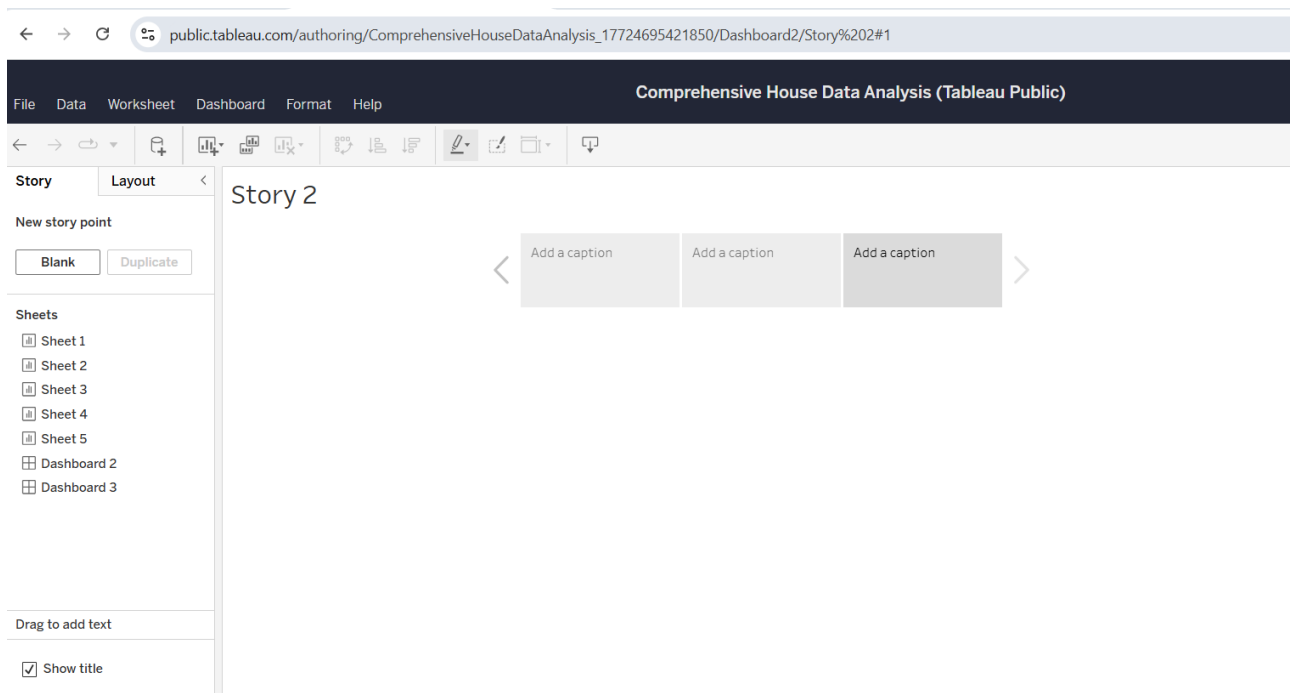
A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

❖ Design a Storyboard

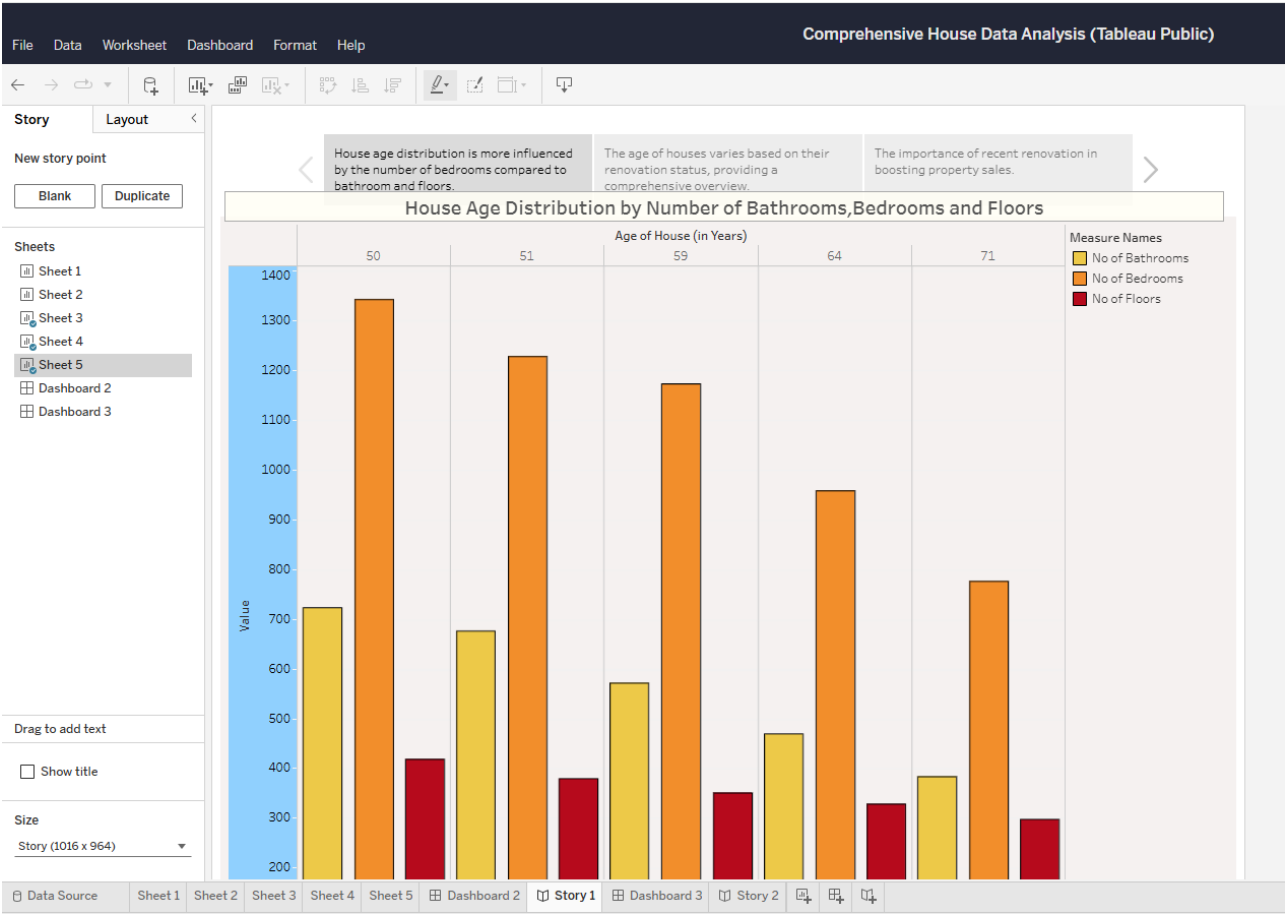
a) Click On **New Story** option.

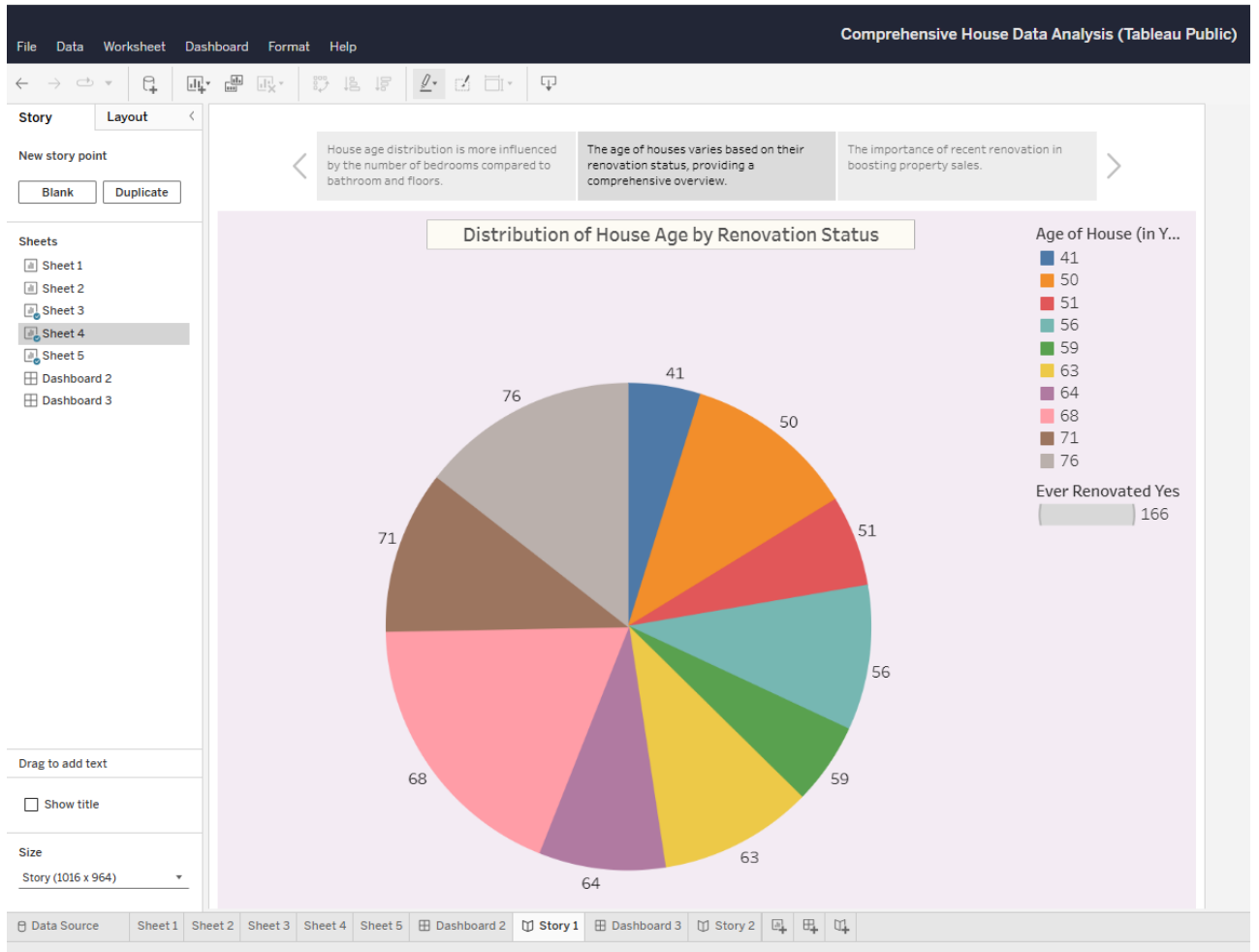


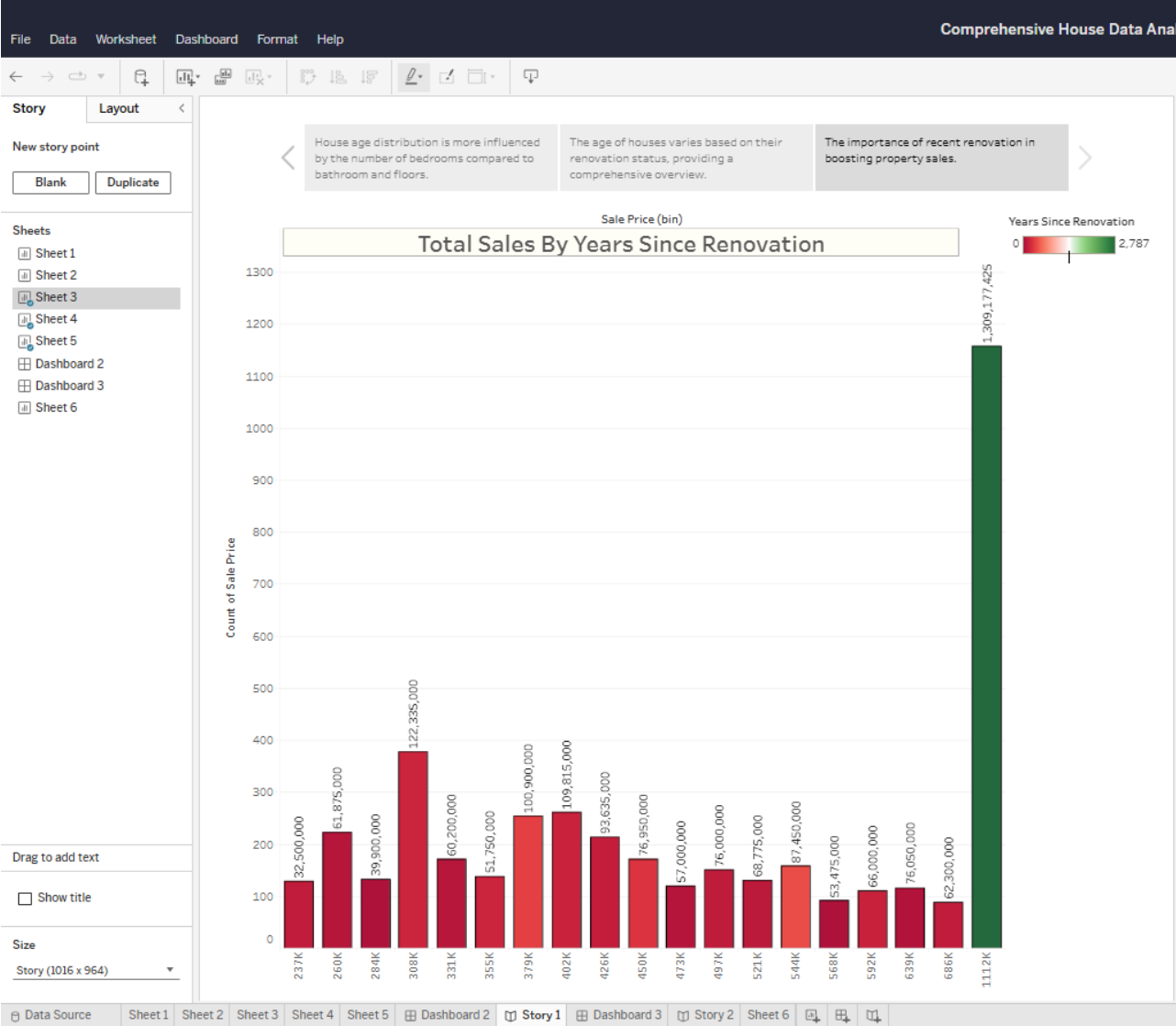
b) Here I have to add 3 sheets into one story so click on **Blank**.



c) Drag and Drop Sheets from Story panel into white Screen and add caption for each sheet.







6) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

❖ Amount of Data Loaded

"Amount of Data Loaded" refers to the quantity or volume of data that has been imported, retrieved, or loaded into a system, software application, database, or any other data storage or processing environment. It's a measure of how much data has been successfully processed and made available for analysis, manipulation, or use within the system.

public.tableau.com/authoring/HouseAgeDistributionbyNumberofBathroomsBedroomsandFloors/Story1#1

House Age Distribution by Number of Bathrooms, Bedrooms and Floors (Tableau Public)

Transformed_Housing_Data2

Extract contains all data, 26-Feb-2026, 8:40:30 pm

Filters 0 Add

Transformed_Housing_D...

Transformed_Housing_Da... 33 fields 21609 rows

100 rows

Transformed_Housing_Data2.csv

No description available.

Type	Field Name	Phys...	Rem...
#	Sale Price	Transf...	Sale...
#	No of Bedroom	Transf...	No of ...
#	No of Bathroom	Transf...	No of ...
#	Flat Area (in Sq	Transf...	Flat Ar...
#	Lot Area (in Sq	Transf...	Lot Ar...
#	No of Floors	Transf...	No of ...

#	Transformed_Housing_Data2.csv	#	Transformed_Housing_Data2.csv	#	Transformed_Housing_Data2.csv	#	Transformed_Housing_Data2.csv	#	Transformed_Housing_Data2.csv
	Sale Price		No of Bedrooms		No of Bathrooms		Flat Area (in Sqft)		Lot Area (in Sqft)
	221900.00		3		1.00000		1.180.00		5,650.00
	538000.00		3		2.25000		2,570.00		7,242.00
	180000.00		2		1.00000		770.00		10,000.00
	604000.00		4		3.00000		1,960.00		5,000.00
	510000.00		3		2.00000		1,680.00		8,080.00
	1129575.00		4		4.50000		5,420.00		101,930.00
	257500.00		3		2.25000		1,715.00		6,819.00
	291850.00		3		1.50000		1,060.00		9,711.00
	229500.00		3		1.00000		1,780.00		7,470.00
	323000.00		3		2.50000		1,890.00		6,560.00

❖ Utilization of Filters

"Utilization of Filters" refers to the application or use of filters within a system, software application, or data processing pipeline to selectively extract, manipulate, or analyse data based on specified criteria or conditions.

1) Selected “Sales Price” as a Top N Filters.

- a) Drage and drop “Sales Price” into Filters Filed.
- b) Right click on “Sales Price” and click on **Edit Filter**.
- c) Then Click on **Top/Bottom**.

Filter [Sale Price] X

General

List

Custom value list ▼

Search or add text Add Search

Clear List ☐ Exclude selected values

☒ Include all values when empty

> Wildcard

> Condition

> Top/Bottom

Summary

Selection: Selected all values

Wildcard: All

Condition: None

Limit: Top 25 by SUM([Age of House (in Years)])

Reset Apply Cancel OK

d) Select **By Field** and add Top 25 records and click OK.

Filter [Sale Price]

> General

> Wildcard

> Condition

> Top/Bottom

None

By field

Rank

Count

Top

Enter a Value...25

Field

Aggregation

Age of House (in Years)

Sum

By formula

Rank

Count

Top

Enter a Value...10

Formula

Summary

Selection:

Selected all values

Wildcard:

All

2) Select “Age of House (in Year)” as a Top N records.

- a) Drage and drop “Age of House (in Year)” into Filters Filed.
- b) Right click on “Sales Price” and click on Edit Filter.
- c) Then Click on Top/Bottom.

Filter [Age of House (in Years)]

General

List

Use all

> Wildcard

> Condition

> Top/Bottom

Summary

Selection: All

Wildcard: All

Condition: None

Limit: Top 10 by SUM([Age of House (in Years)])

Reset

Apply

Cancel

OK

d) Select **By Field** and add Top 10 records and click OK.

Filter [Age of House (in Years)]

> General

> Wildcard

> Condition

> Top/Bottom

None

By field

Rank

Count

Top

Enter a Value...

10

Field

Aggregation

Age of House (in Years)

Sum

By formula

Rank

Count

Top

Enter a Value...

10

Formula

Summary

Selection:

All

3) House Age Distribution by Top N Filters.

- a) Drage and drop “Age of House (in Year)” into Filters Filed.
- b) Right click on “Sales Price” and click on **Edit Filter**.
- c) Then Click on **Top/Bottom**.

Filter [Age of House (in Years)]

General

List

Use all

Wildcard

Condition

Top/Bottom

Summary

Selection: All

Wildcard: All

Condition: None

Limit: Top 10 by SUM([Age of House (in Years)])

Reset

Apply

Cancel

OK

d) Select **By Field** and add Top 5 records and click OK.

Filter [Age of House (in Years)]

> General

> Wildcard

> Condition

< Top/Bottom

None

By field

Rank

Count

Top

Enter a Value...

5

Field

Aggregation

Age of House (in Years)

Sum

By formula

Rank

Count

Top

Enter a Value...

10

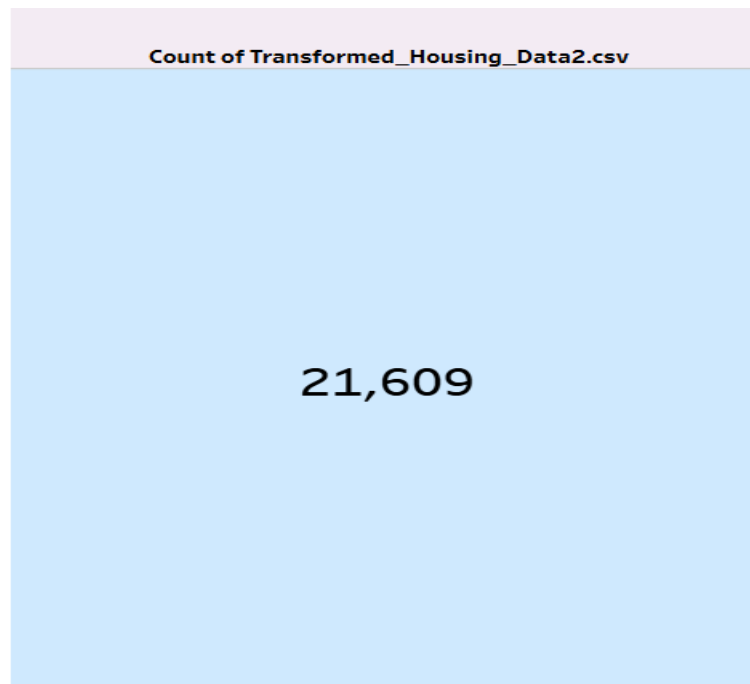
Formula

Summary

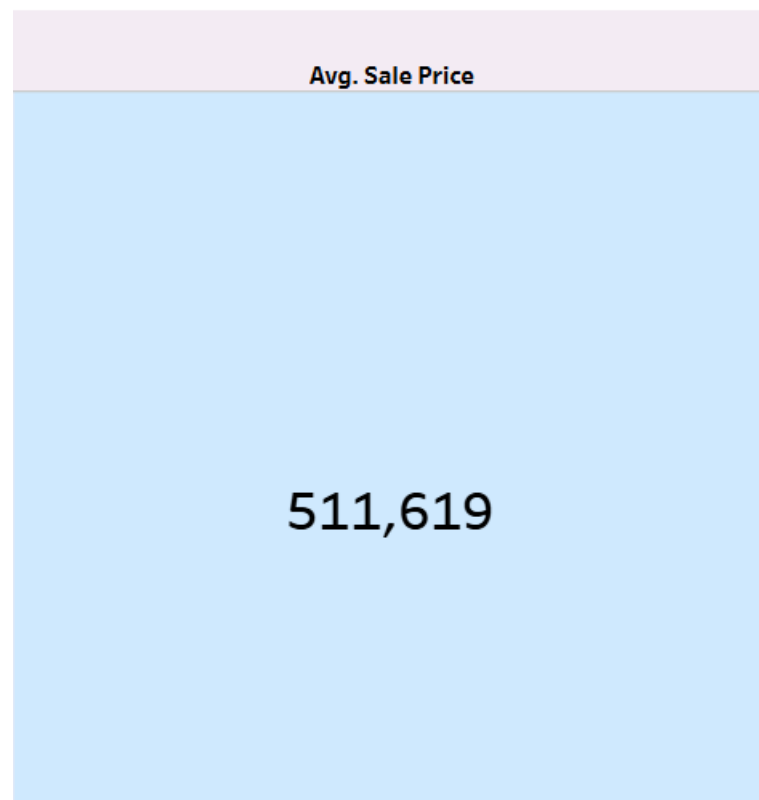
Selection: All

❖ No of Visualizations/ Graphs

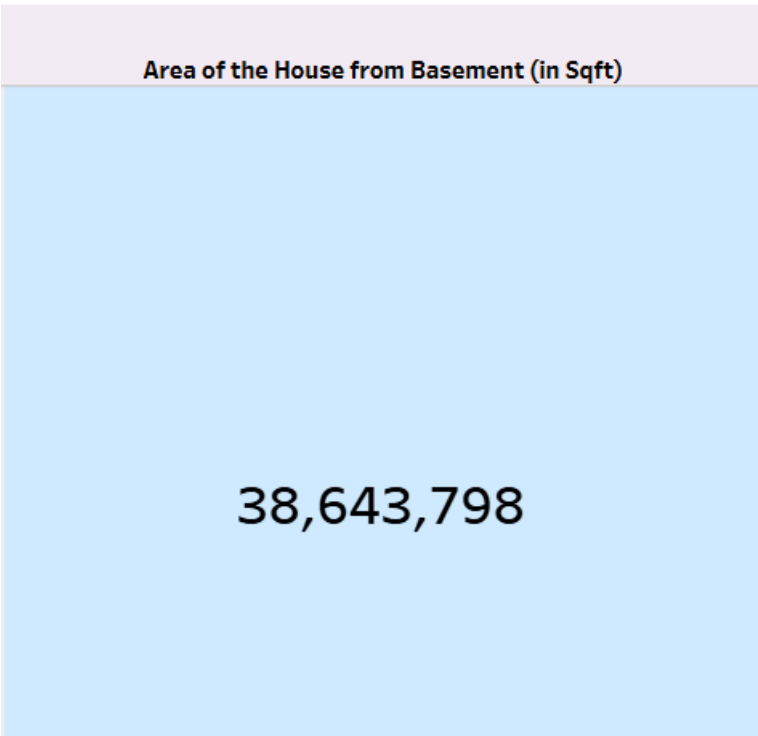
1) Count of Transformed Housing Data



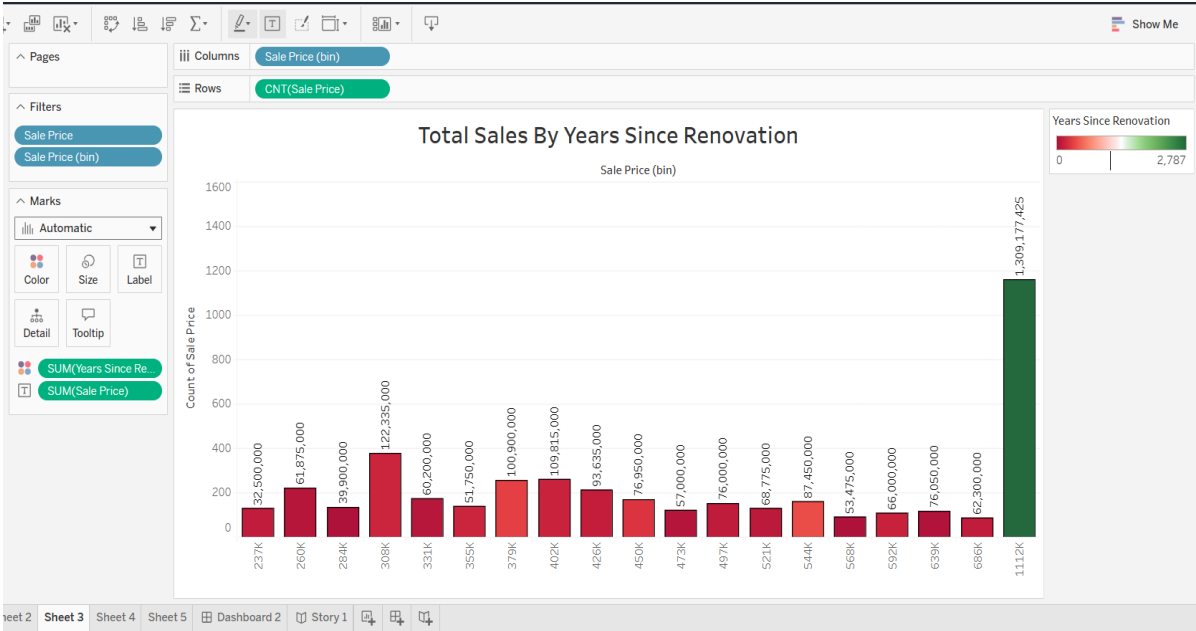
2) Average Sale Prices



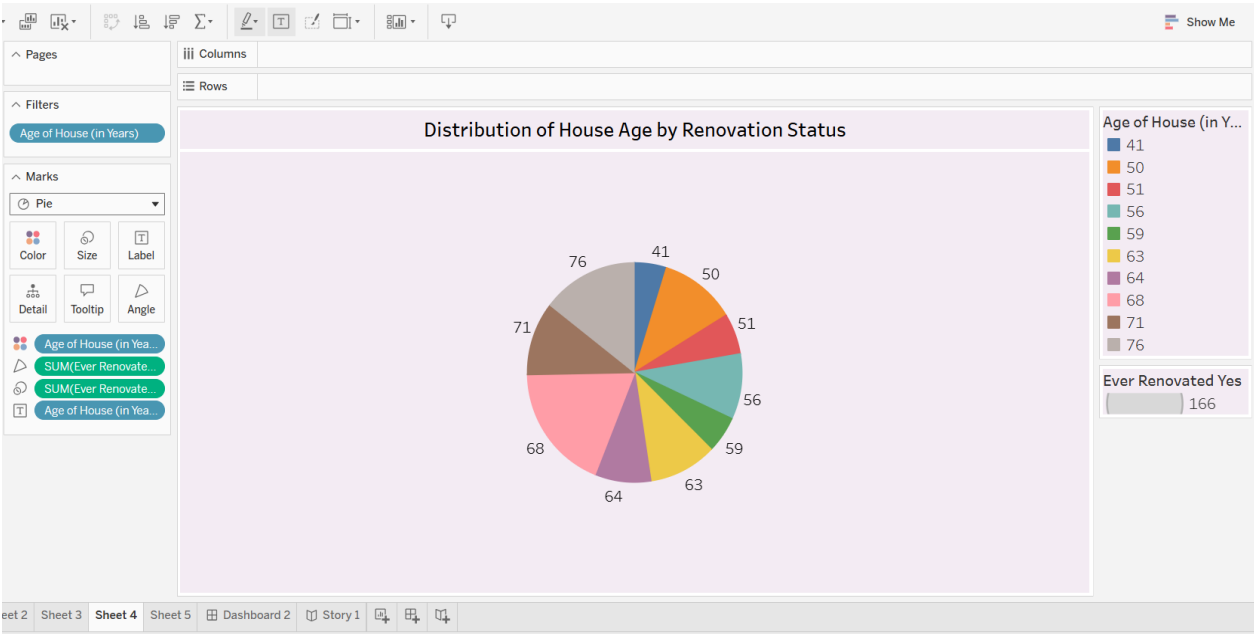
3) Area of House from Basement (in Sqft)



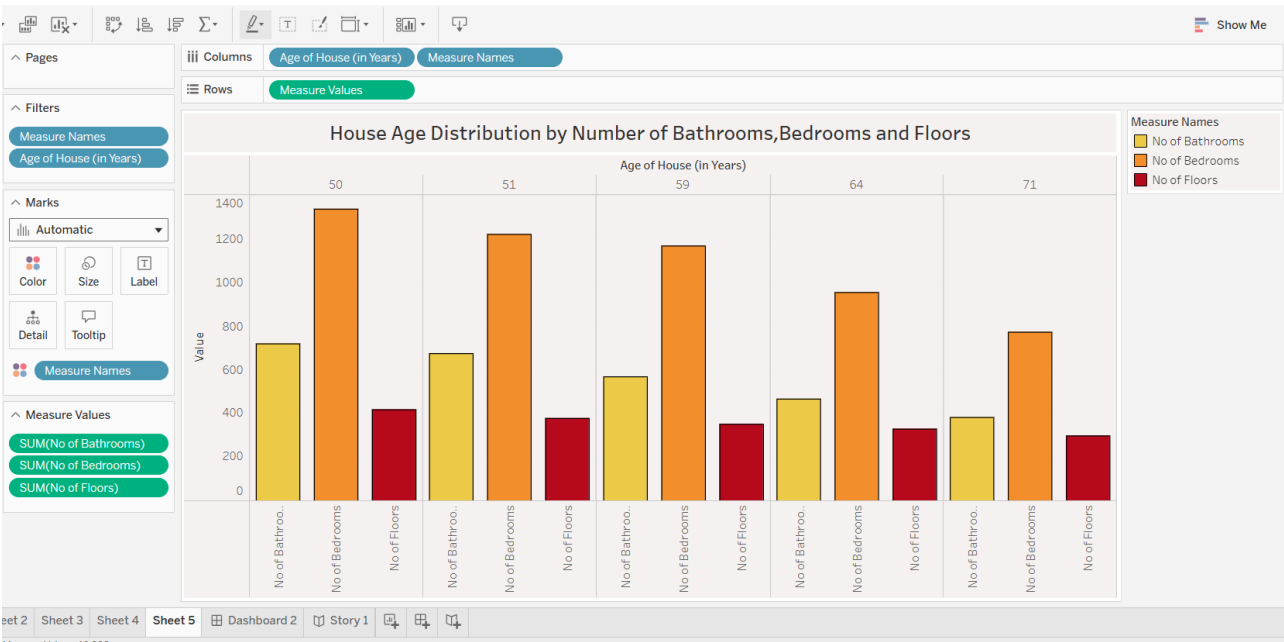
4) Total Sales by Years Since Renovation



5) Distribution of House Age by Renovation Status



6) House Age Distribution by Number of Bathrooms, Bedrooms, and Floors



7) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

❖ Publishing

a) Sign in to Tableau Public

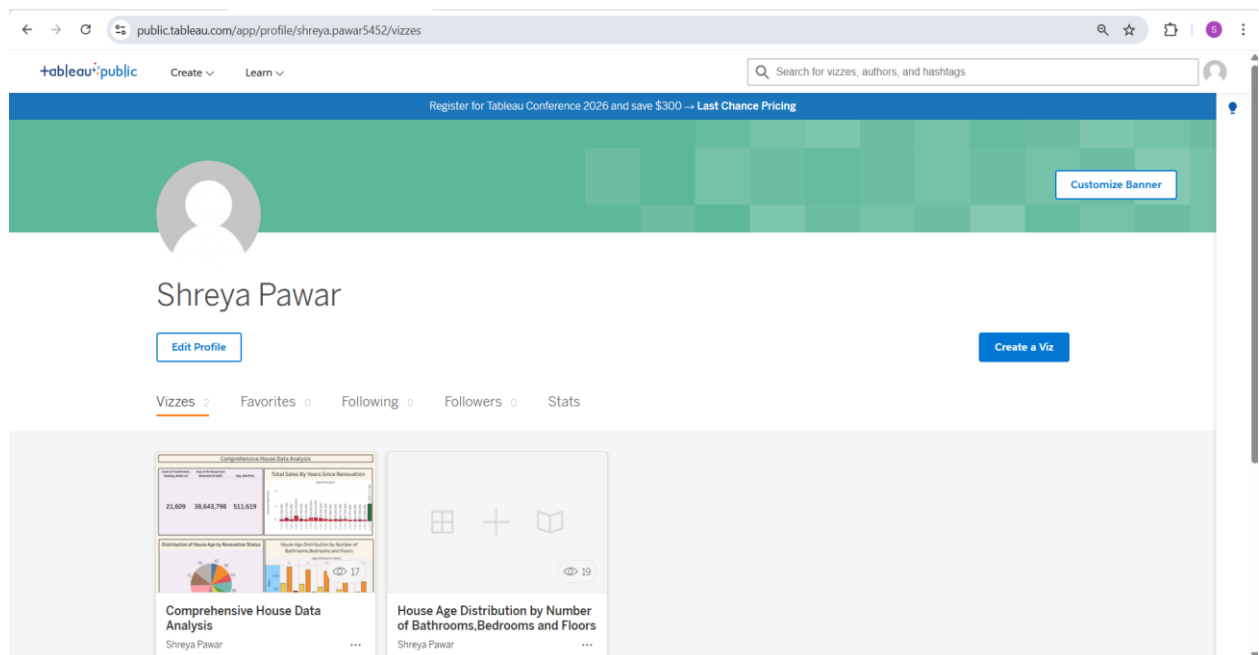
In Tableau Desktop, go to **File > Save to Tableau Public**.

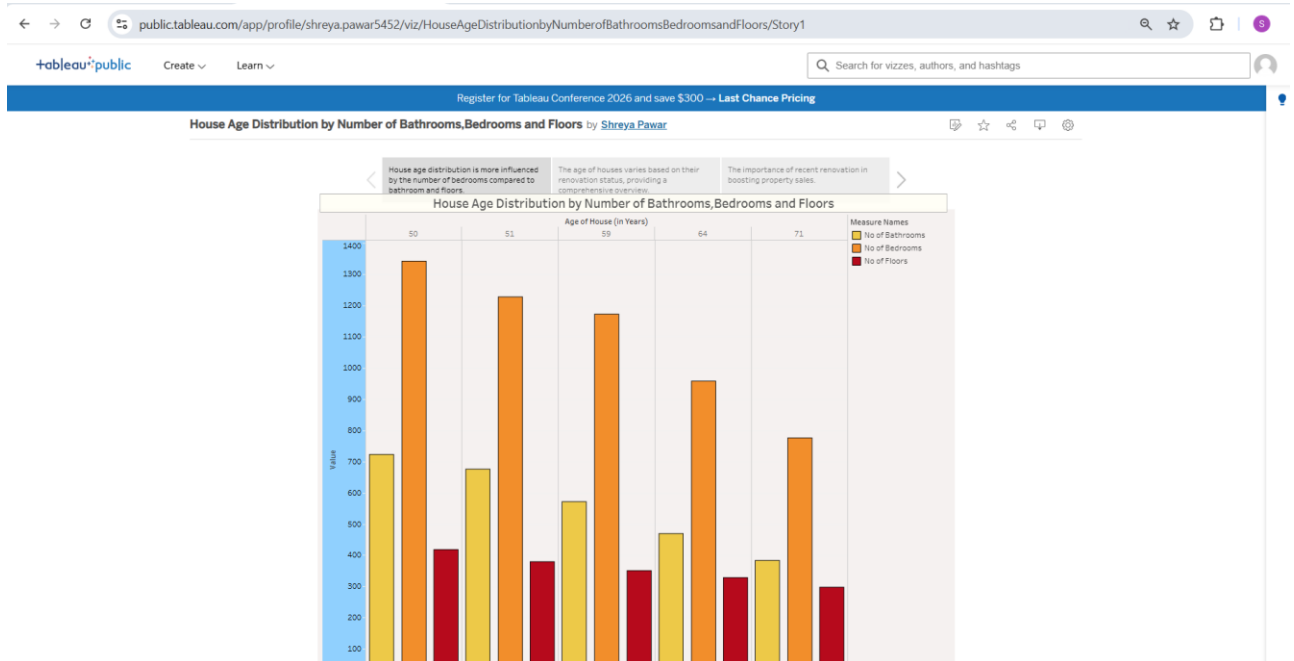
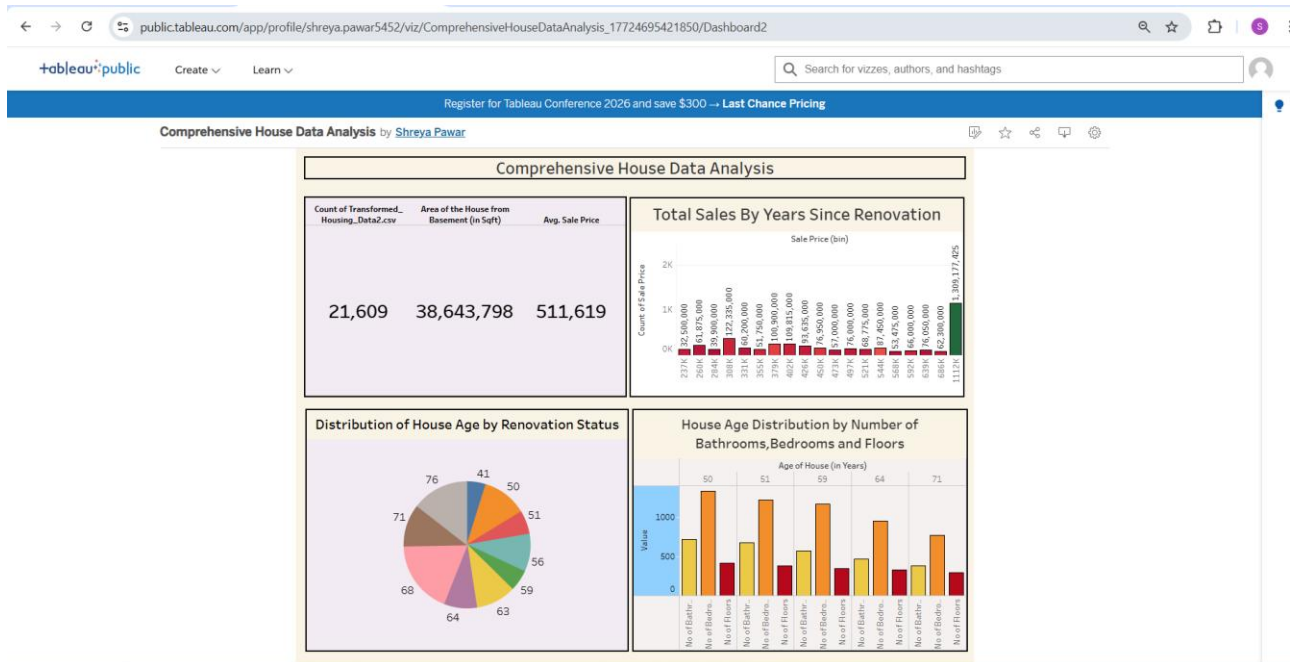
If you're not already signed in, a login window will appear.

Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) Save and Publish

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.

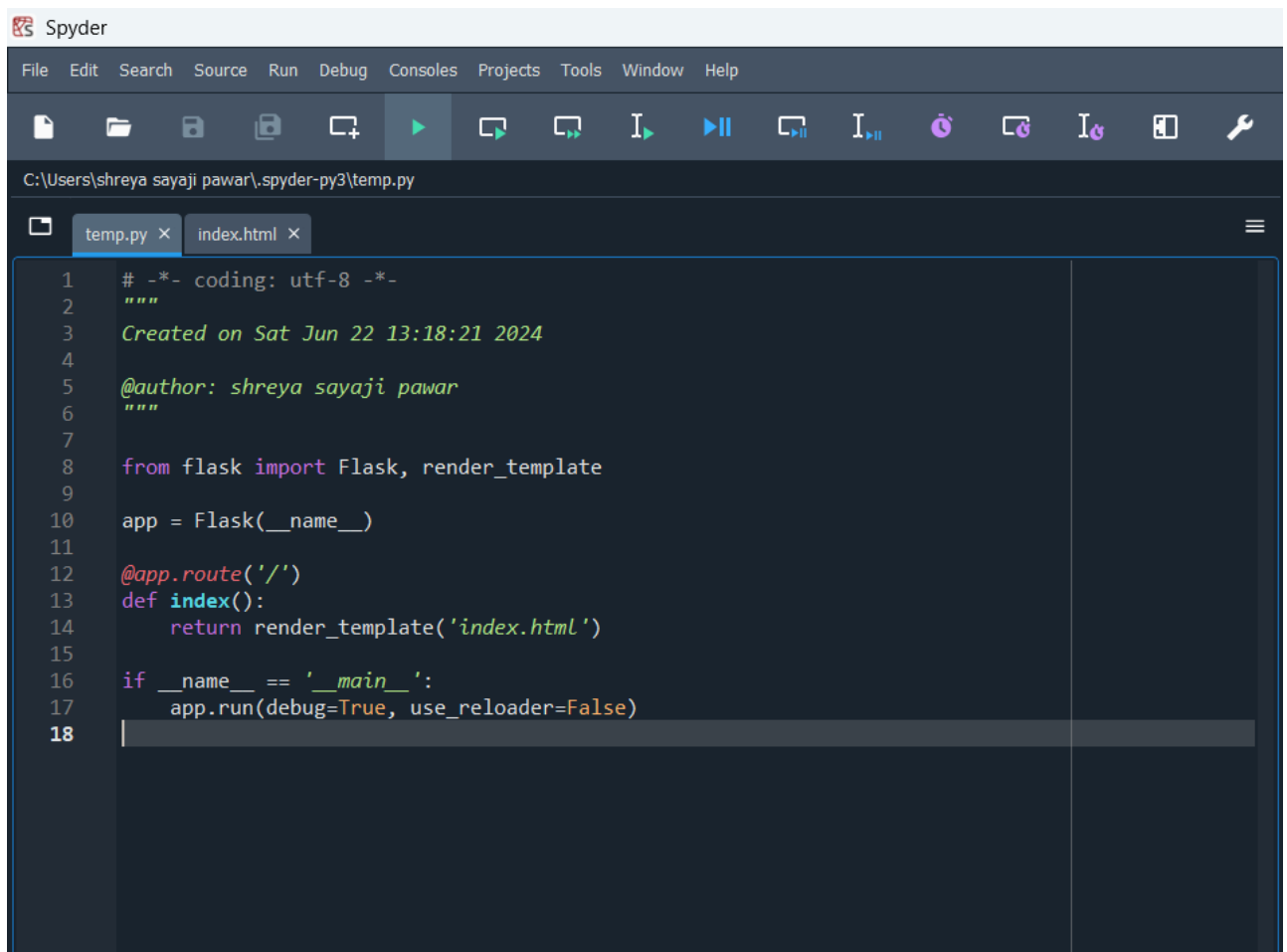




❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

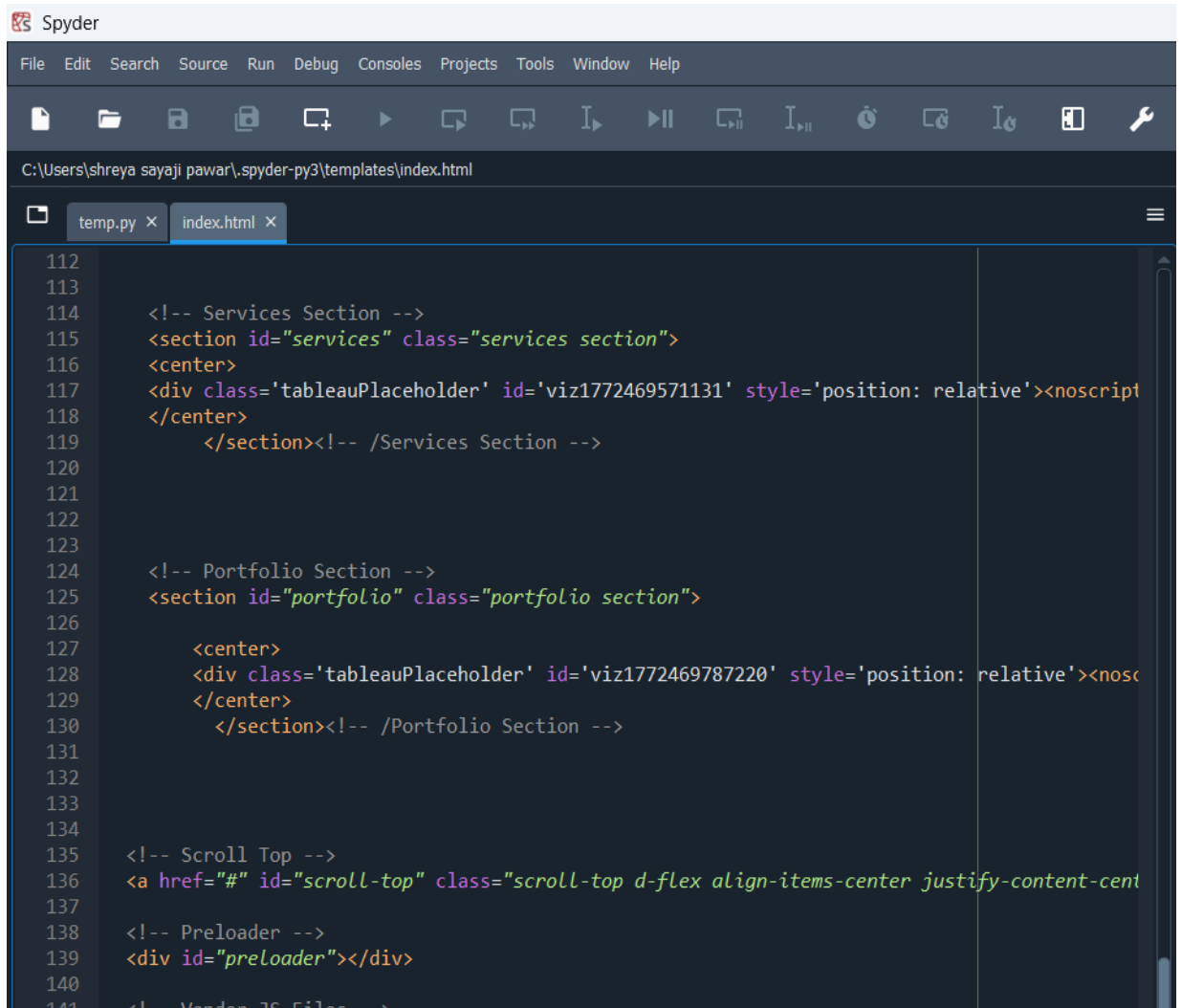
1) temp.py



```
1  # -*- coding: utf-8 -*-
2  """
3  Created on Sat Jun 22 13:18:21 2024
4
5  @author: shreya sayaji pawar
6  """
7
8  from flask import Flask, render_template
9
10 app = Flask(__name__)
11
12 @app.route('/')
13 def index():
14     return render_template('index.html')
15
16 if __name__ == '__main__':
17     app.run(debug=True, use_reloader=False)
18
```

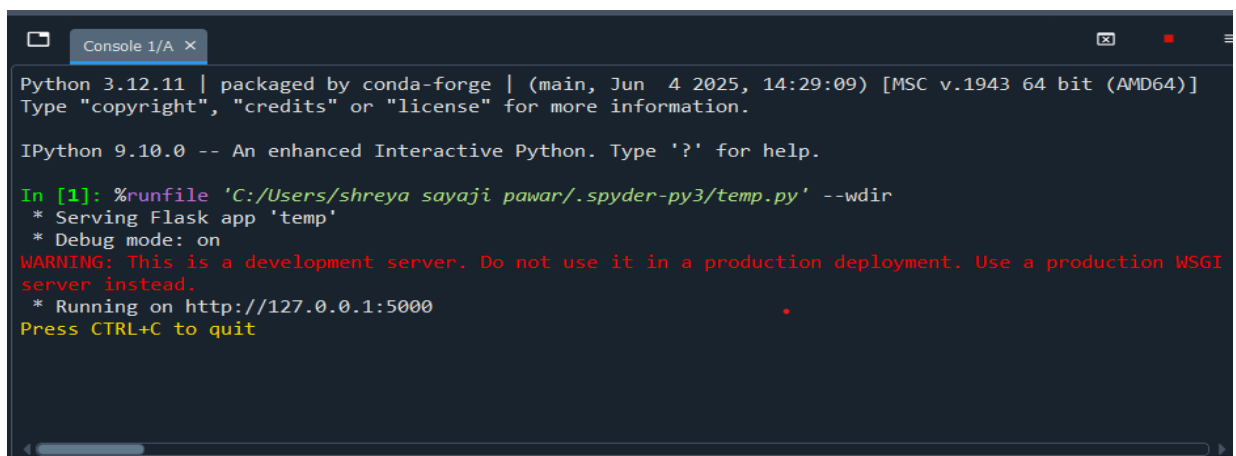
2) Adding Dashboard and Story on HTML page.

- Open Tableau Public profile and open Dashboard/Story.
- Click on share option and copy the embed code.
- Paste in HTML page



```
112
113
114 <!-- Services Section -->
115 <section id="services" class="services section">
116 <center>
117 <div class='tableauPlaceholder' id='viz1772469571131' style='position: relative'><noscript>
118 </center>
119 </section><!-- /Services Section -->
120
121
122
123
124 <!-- Portfolio Section -->
125 <section id="portfolio" class="portfolio section">
126
127 <center>
128 <div class='tableauPlaceholder' id='viz1772469787220' style='position: relative'><noscript>
129 </center>
130 </section><!-- /Portfolio Section -->
131
132
133
134
135 <!-- Scroll Top -->
136 <a href="#" id="scroll-top" class="scroll-top d-flex align-items-center justify-content-cent
137
138 <!-- Preloader -->
139 <div id="preloader"></div>
140
141 <!-- Vendor JS Files -->
```

- Run the temp.py and copy the port number and paste it on Browser.



```
Python 3.12.11 | packaged by conda-forge | (main, Jun 4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.10.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile 'C:/Users/shreya sayaji pawar/.spyder-py3/temp.py' --wdir
* Serving Flask app 'temp'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

VISUALIZING HOUSING MARKET TRENDS: AN ANALYSIS OF SALE PRICES AND FEATURES

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau



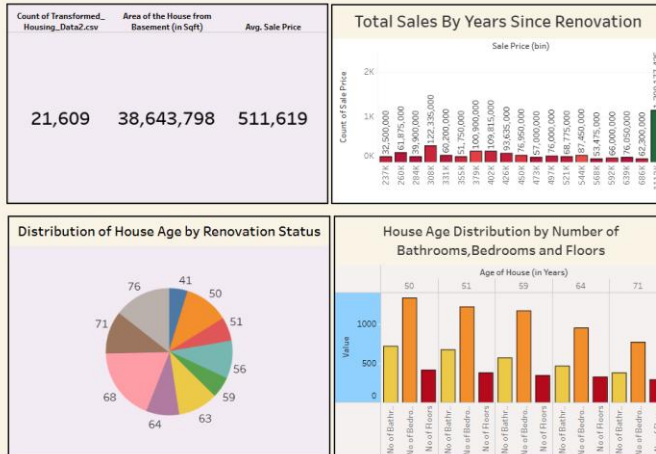
This project focuses on analyzing housing market trends by examining property features and sale prices using Tableau. The dataset was cleaned and prepared to ensure accuracy, including verification of data types, removal of null values, and validation of total records loaded. Key metrics such as total properties, average sale price, and basement area were calculated to gain meaningful insights.

- ✓ Analyzed housing market data to understand pricing trends and property features.
- ✓ Cleaned and prepared the dataset for accurate visualization in Tableau.
- ✓ Created interactive dashboards to explore sales performance and housing characteristics.

Interactive dashboards were created to visualize relationships between price, location, property size, and other housing characteristics. The analysis helps identify pricing patterns and performance trends within the housing market, making the data easy to understand and interpret through clear and structured visualizations.



Comprehensive House Data Analysis

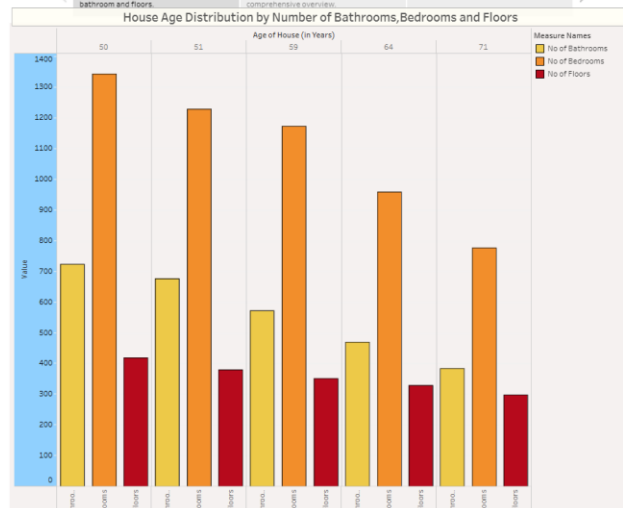


View on Tableau Public

House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.

The age of houses varies based on their renovation status, providing a comprehensive overview.

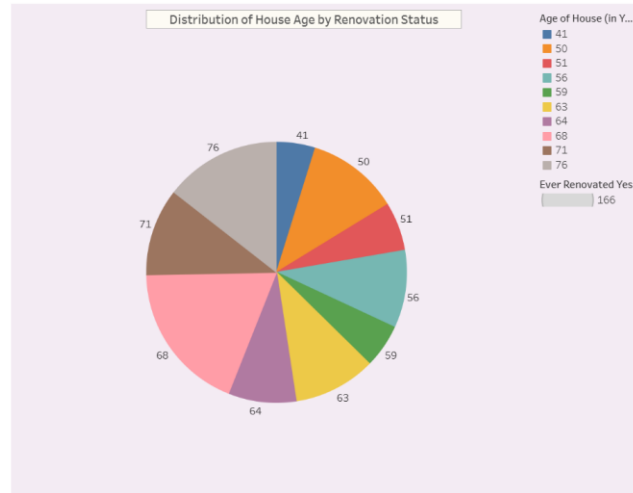
The importance of recent renovation in boosting property sales.



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